CFD interpretation practical: transonic airfoil screening

Instructions and reference material

You have **30 minutes**. Work individually. No calculator, network access, AI tool, code, or notes are permitted. Show a short calculation where requested; concise evidence-based statements are sufficient elsewhere.

Scenario

A 625 kg high-subsonic cargo UAV, including its payload, is screened at one flight condition. The results below come from a two-dimensional inviscid finite-volume airfoil calculation. Three 12%-thick candidate configurations have different camber and incidence.

Reference relations

Use $g = 10$ N/kg and the supplied load factor $n = 1.2$:

$$W = mg, \qquad L_{\text{req}} = nW, \qquad L = q_\infty S C_L, \qquad D = q_\infty S C_D.$$

For this problem, $q_\infty S = 10.0$ kN. The strong-shock screening limit is

$$M_{\text{surface,max}} \leq 1.60.$$

A rapid decrease in Mach number paired with a sharp pressure recovery in a surface C_p curve indicates a shock. A strong shock can raise transonic wave drag and may cause separation in a real viscous flow.
Model boundary. The CFD coefficients are used here as a simplified wing-screening approximation. They do not certify viscous drag, separation, three-dimensional effects, structural loads, stability, or full aircraft performance.

1. Requirements and preliminary recommendations [10 points]

Candidate	Camber	α	C_L	C_D	$M_{\text{surface,max}}$
A: NACA 4412	high	2.25°	0.958	0.166	1.70
B: NACA 2412	moderate	1.75°	0.799	0.110	1.56
C: NACA 1412	low	1.50°	0.627	0.079	1.46

Engineer A: "Select A. It has the greatest lift coefficient and therefore the greatest payload margin."
Engineer B: "Select C. It has the smallest drag coefficient and therefore is the most efficient choice."

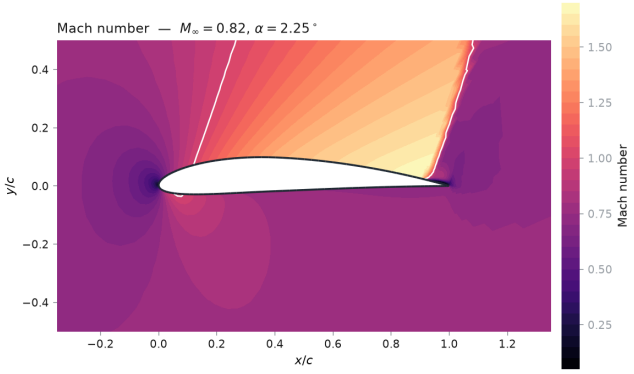
1. (2 points) Calculate L_{req} and $C_{L,\text{req}}$.

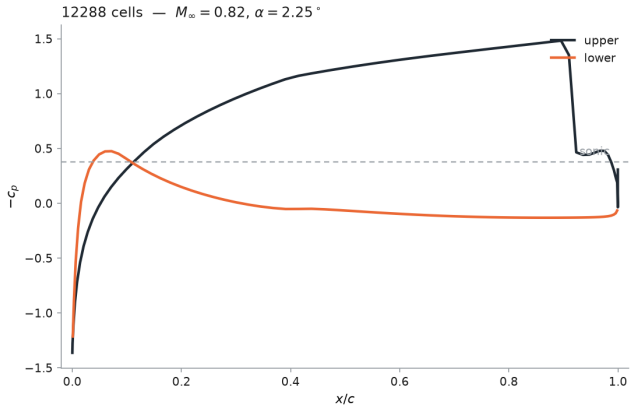
2. (2 points) Calculate Candidate B's predicted lift and drag.

3. (4 points) For each candidate, classify it as *satisfies both requirements*, *fails lift only*, *fails the strong-shock limit only*, or *fails both*. Give one numerical reason for each classification.

4. (2 points) In one sentence each, state why Engineer A's and Engineer B's recommendations are incomplete.

2. Flow-field evidence [7 points]





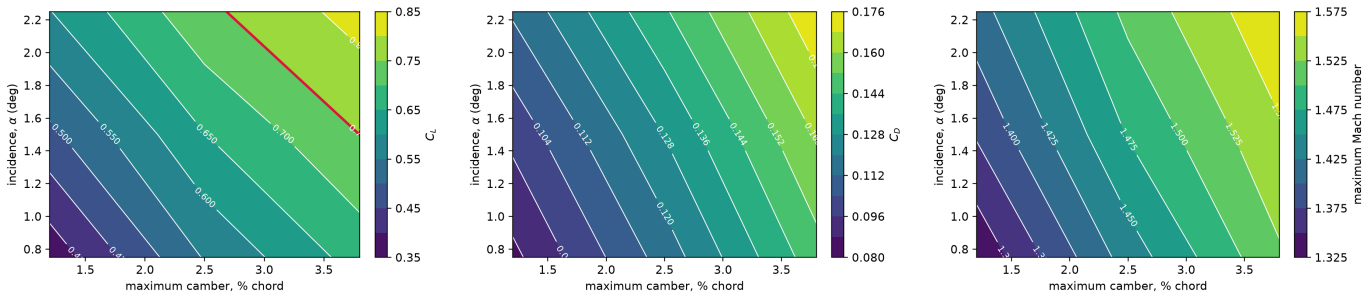
Candidate A: NACA 4412 at $M_\infty = 0.82$ and $\alpha = 2.25°$. These are the custom course results used in this practical.

5. (2 points) On the supplied Mach and C_p plots, mark the shock location.

6. (3 points) Explain how increasing aerodynamic loading can increase lift while also producing a stronger shock and higher drag. Cite one observation from a Mach contour and one observation from a C_p curve.

7. (2 points) Explain why Candidate A must be rejected despite its large C_L.

3. JAX sweep and design recommendation [9 points]



Custom JAX-adjoint screening sweep: fixed 12% thickness and camber crest at $x/c = 0.40$. The black solid line marks $C_L = 0.75$; the black dashed line marks $M_{\max} = 1.60$. Dark-to-light shading uses the colorblind-safe cividis scale.

8. (2 points) Identify the feasible region of the (m, α) sweep: it must meet both the lift and strong-shock requirements. Why is the upper-right portion of the sweep not automatically the best design region?

9. (2 points) The supplied custom results use one mesh. State the next numerical check required before treating the drag ranking as reliable, and explain why.

10. (5 points) Use all three maps to choose an approximate (m, α) point that minimizes C_D while meeting $C_L \geq 0.75$ and $M_{\max} \leq 1.60$. State approximate C_L, C_D, and $M_{\max}$ values at that point; explain the trade-off with a nearby equal-lift alternative; and state the next simulation or experiment, with its purpose.

Answer key (instructor review)

These are the expected conclusions. Award credit for concise, physically supported equivalents.

1. $W = 625(10) = 6250$ N; $L_{\text{req}} = 1.2(6250) = 7500$ N $= 7.5$ kN; therefore $C_{L,\text{req}} = 7.5/10.0 = 0.75$.

2. For B, $L = 10.0(0.799) = 7.99$ kN ≈ 8.0 kN and $D = 10.0(0.110) = 1.10$ kN.

3. A fails the strong-shock limit only: $C_L = 0.958 > 0.75$, but $M_{\max} = 1.70 > 1.60$. B satisfies both: $C_L = 0.799 > 0.75$ and $M_{\max} = 1.56 < 1.60$. C fails lift only: $C_L = 0.627 < 0.75$, although $M_{\max} = 1.46 < 1.60$.

4. Engineer A ignores the shock constraint and the associated wave-drag risk. Engineer B ignores the required payload lift: C does not meet it.

5. Mark the abrupt upper-surface Mach decrease near the trailing portion of the airfoil (approximately $x/c = 0.9$–1.0), paired with the sharp C_p recovery.

6. Greater camber/incidence increases circulation and upper-surface acceleration, raising lift. The high-Mach region over the upper surface then recompresses abruptly near the shock; the C_p curve shows the corresponding sharp recovery. This creates wave-drag risk (and, in real viscous flow, possible separation).

7. A produces adequate lift but violates $M_{\max} \leq 1.60$: its value is 1.70, indicating an unacceptably strong shock by this screen.

8. The feasible region is the intersection of the $C_L \geq 0.75$ side of the solid contour and the $M_{\max} \leq 1.60$ side of the dashed contour. The upper right is not automatically best because increasing camber and incidence raises drag and eventually violates the shock constraint.

9. Perform a grid-refinement (mesh-convergence) study, comparing C_L, C_D, and shock location/strength over progressively finer meshes. Drag and shocks can be sensitive to numerical resolution, so one mesh cannot establish a reliable ranking.

10. Accept a point on the minimum-C_D portion of the feasible boundary, supported by approximate readings from all three maps. It should meet both constraints; a higher-camber/lower-incidence equal-lift alternative generally has different drag and shock margin. The next step should be, for example, a mesh-refined viscous CFD calculation (or a wind-tunnel test) to assess drag, shock/boundary-layer interaction, and separation before a design decision. *After regenerating the maps, replace this sentence with the selected numerical reference point and tolerances for grading.*